



DEPARTMENT OF
PLANT SCIENCE &
LANDSCAPE ARCHITECTURE

November 25, 2025

Dear Dr. Lin & Dr. Foyer,

Please consider this manuscript: "Stark sexual dimorphism in flowering responses to vernalization in a monoecious shrub" as a *Brief Communication* in *Plant, Cell & Environment*. My study investigates how male and female flowers of the monoecious shrub *Comptonia peregrina* respond to key environmental cues that regulate the competency to flower.

Temperate woody plant species rely on vernalization (winter chilling), forcing (spring warming), and photoperiod to coordinate flowering. and it has been generally assumed that both male and female sex phases should respond similarly to these environmental cues. Using data from a controlled-environment experiment that independently manipulated these cues, I found that this assumption does not hold for *C. peregrina*. Male catkins exhibited a strong requirement for extended vernalization—virtually no catkins opened after a shorter vernalization period (30 days), whereas a longer vernalization (60 days) promoted flowering regardless of forcing temperature or daylength. In contrast, female flowers opened under both short and long vernalization treatments.

This study demonstrates that male and female reproductive phases—even within a single individual—can have significantly different environmental thresholds. I believe these findings will be of great interest to readers of *Plant, Cell & Environment*, especially researchers studying environmental control of plant reproduction, including those working on phenology, pollination, and the integration of temperature and photoperiod cues. Because this study reveals distinct vernalization responses for male and female sex phases, it also connects broader questions about how plants coordinate life-history transitions across organs, and how these processes may become decoupled with continued climate warming.

As this work deals broadly with developmental transitions in woody plants, I recommend Dr. Kate McCulloh or Dr. Sanna Sevanto to review this manuscript, as well as Dr. Yingying Xie (xiey2@nku.edu) who is an expert on sexual dimorphism in plant phenology.

This manuscript is 785 words in length. contains one figure, and is not under consideration elsewhere. I hope that you will find it suitable for publication in *Plant Cell & Environment* and look forward to hearing from you.

Sincerely,

A handwritten signature in black ink, appearing to read "Daniel Buonaiuto".

Daniel Buonaiuto